CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

# Chapter 2: Banking (Recurring Deposit Accounts)

Topic-wise short notes with solved examples

In a recurring deposit account a fixed sum is deposited every month for a fixed period. This chapter is about finding the interest and the maturity value, and working backwards from them.

## 1 The recurring deposit account

- In a **recurring deposit (R.D.) account**, the account holder deposits the same amount every month for a fixed number of months.
- At the end of the period the bank pays back the total amount deposited together with the interest. This payment is the **maturity value (M.V.)**.
- Symbols: $P$ = monthly deposit (instalment), $n$ = number of months, $r$ = rate of interest per cent per annum.
- $n$ is always in **months**. Change years to months first: 2 years $= 24$ months, $1\frac{1}{2}$ years $= 18$ months.

## 2 Interest and maturity value

- The interest on a recurring deposit is

$$I = P \times \frac{n(n+1)}{2 \times 12} \times \frac{r}{100}$$

- Total amount deposited $= P \times n$.

$$\text{M.V.} = P \times n + I$$

- Where the formula comes from: the first deposit earns interest for $n$ months, the second for $n-1$ months, and so on down to 1 month. The total is $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ months, which is $\frac{n(n+1)}{2 \times 12}$ years of interest on a single deposit $P$.

**Example 2.1** Mohan deposits ₹600 per month in a recurring deposit account for 20 months at 10% per annum. Find the interest and the maturity value.

SOLUTION

$P = 600, n = 20, r = 10$

$$I = 600 \times \frac{20 \times 21}{2 \times 12} \times \frac{10}{100} = 600 \times 17.5 \times 0.1 = ₹1{,}050$$

Total deposited $= 600 \times 20 = ₹12{,}000$

**Interest ₹1,050; maturity value ₹13,050**

## 3 Finding the monthly deposit

- Put the given maturity value in $\text{M.V.} = Pn + I$ with $P$ as the unknown.

**Example 3.1** Ritu has a recurring deposit account for 2 years at 8% per annum. If she receives ₹26,000 at maturity, find her monthly deposit.

SOLUTION

$n = 24, r = 8.\ I = P \times \frac{24 \times 25}{2 \times 12} \times \frac{8}{100} = P \times 25 \times \frac{8}{100} = 2P$

$\text{M.V.} = 24P + 2P = 26P = 26000$

**Monthly deposit = ₹1,000**

## 4 Finding the rate of interest

- Find the interest as M.V. − total deposited, then use the interest formula with $r$ as the unknown.

**Example 4.1** Amit deposits ₹200 per month for 36 months in a recurring deposit account and receives ₹8,310 at maturity. Find the rate of interest.

SOLUTION

Total deposited $= 200 \times 36 =$ ₹7,200, so $I = 8310 - 7200 =$ ₹1,110

$1110 = 200 \times \frac{36 \times 37}{2 \times 12} \times \frac{r}{100} = 200 \times 55.5 \times \frac{r}{100} = 111r$

**$r = 10\%$ per annum**

## 5 Finding the time

- With $n$ unknown, the interest formula gives a quadratic equation in $n$. Solve it and take the positive whole-number root.

**Example 5.1** Kiran deposits ₹300 per month in a recurring deposit account at 12% per annum and earns ₹900 as interest. Find the time for which the account was held.

SOLUTION

$900 = 300 \times \frac{n(n+1)}{2 \times 12} \times \frac{12}{100} = 1.5\, n(n+1)$

$n(n+1) = 600$, so $n^2 + n - 600 = 0$ and $(n + 25)(n - 24) = 0$

$n = 24$ (the negative value is rejected)

**The account was held for 24 months, that is 2 years.**

**REMEMBER**

$n(n+1)$ is a product of two consecutive numbers. Look for the two consecutive numbers that give the product: $24 \times 25 = 600$.

## 6 Common mistakes

- Using $n$ in years. The formula needs months.
- Forgetting the 12 in the denominator: the formula has $2 \times 12 = 24$.
- Giving the interest when the maturity value is asked, or the other way round.
- Adding $P$ instead of $P \times n$ to the interest.

MathSetu